AIRTEL SYNC 2026 · DIGITAL TRACK

Udyam Digital

A mobile-friendly website and WhatsApp journey that takes a small business from one sentence to a booked installation, in its own language. Supporting material for our 7-slide submission.

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LINKS

Demo video
2 min 58 s
drive.google.com · Demo-Airtel.mp4

Source code
GitHub, public
github.com/k9927/udyam-digital

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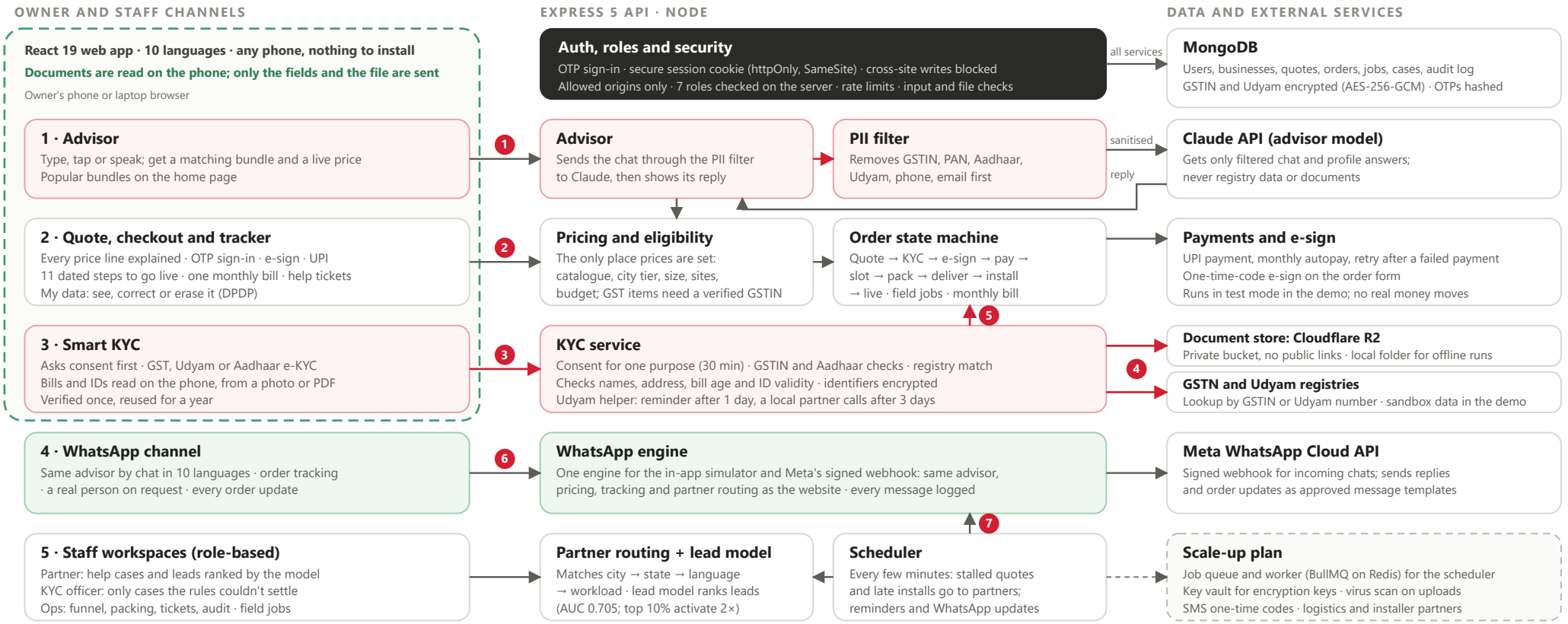
RUN IT IN 5 MINUTES

```
git clone https://github.com/k9927/udyam-digital.git
cd udyam-digital && npm run install:all
cp backend/.env.example backend/.env
npm run dev:api        # terminal 1
npm run dev:web        # terminal 2, then open http://localhost:5173
```

Needs Node 20+ and MongoDB. Demo staff, sample businesses and the 12 specimen applicants load on first start; **Demo: view as…** switches between all seven roles. `npm test` runs the 78 automated tests.

Project Udyam Digital: solution architecture

Owners come in by website or WhatsApp; every staff role has its own workspace; one Express API serves them all. Dashed boxes are the scale-up plan.



DATA FLOWS

1. Chat passes the PII filter before it reaches Claude; Claude's reply returns to the advisor, but only the pricing engine sets products and prices.
2. An accepted quote becomes an order: the owner e-signs, pays by UPI and books an installation slot.
3. After consent, the phone reads the document and sends the fields and the file; the server checks the file is genuine before saving it.
4. Files are kept in a private R2 bucket and opened only through the API, with every staff view logged; IDs are checked against the registries.
5. Once the business is verified, waiting orders move on and GST-only products unlock before the owner signs.
6. WhatsApp messages use the same advisor, pricing, tracking and partner routing as the website.
7. The scheduler opens partner cases and sends reminders and WhatsApp updates; at scale, a job queue takes this over.

KEY

- Built
- Consent and personal-data path
- WhatsApp channel
- Scale-up plan
- Personal-data flow
- Owner's device
- Scale-up link

Built with MongoDB, Express 5, React 19 and Node 24 · on-device OCR (tesseract.js, pdf.js) · Claude API · Cloudflare R2 · 78 automated tests · synthetic case data.

Udyam Digital: data privacy and security note

Team NATPAT · TIET, Patiala · Airtel SYNC 2026

How the journey applies the DPDP Act, 2023: collect only what each step needs, ask before you look, keep it only as long as needed, show the owner what happened. **BUILT** marks what Udyam Digital does today; **AT LAUNCH** marks the steps a live rollout adds.

1 · WHAT EACH STEP COLLECTS, AND WHY

Step	Data	Purpose	Where it goes, who sees it
Advisor (web or WhatsApp)	Business type, size, sites, city, GST yes/no, goal, optional spend. No ID numbers asked	Recommend an eligible bundle	Claude, with GSTIN, PAN, Aadhaar, Udyam, phone and email removed first; WhatsApp threads kept by number
Sign-in	Mobile number, name; one-time code	Account and updates	Code stored only as a hash and expires automatically
Registry check	GSTIN or Udyam number (a proprietor's GSTIN embeds their PAN)	Confirm the business is real and active	Only after consent; encrypted at rest, always shown masked
Owner e-KYC (no GST or Udyam)	Aadhaar number, typed once (the check digit only catches typos). Verified by OTP through an authorised e-KYC partner (synthetic numbers in the demo)	Verify the owner of an unregistered business	Number discarded; only the result and last 4 digits kept, as on UIDAI's masked Aadhaar
Documents	Bill, signatory ID; for unregistered shops a licence or shop photo (image or PDF), read on the phone	Check names, address, dates, validity	Private bucket, no public link; only the owner and KYC/ops staff; every view logged
Order and field visit	Order, slot, address, delivery and install codes	Deliver, install, bill, support	Codes encrypted; field staff see a masked phone
Partner help	Case reason, city, language, owner's number	Call the owner who needs help	Full number only to the assigned partner; masked elsewhere

2 · SECURITY CONTROLS

- **BUILT** GSTIN and Udyam numbers encrypted with AES-256-GCM and looked up by keyed hash; Aadhaar numbers discarded after e-KYC; delivery and install codes encrypted; one-time codes hashed.
- **BUILT** OTP sign-in, an httpOnly session cookie (secure in production) and 7 roles. Each role reaches only its own screens and data, enforced on the server.
- **BUILT** The server recalculates every price and rule; inputs schema-validated; uploads JPEG, PNG, WebP or PDF up to 10 MB; rate limits on sign-in, Aadhaar OTP and WhatsApp.
- **BUILT** Audit log of consent, registry lookups, document views, KYC decisions and data requests. WhatsApp webhooks are signature-checked.
- **BUILT** Production mode refuses default secrets and turns off demo shortcuts (role switcher, on-screen codes).
- **AT LAUNCH** Keys in a managed vault; TLS for all traffic; signatory liveness check; Aadhaar e-sign; penetration test.

3 · DATA MINIMISATION BY DESIGN

- **BUILT** Registry first; documents only if still needed; verified once, reused for a year. Owners without GST are steered to free Udyam first; Aadhaar e-KYC is the fallback.
- **BUILT** Saying "we have GST" never unlocks GST-only products; only a verified GSTIN does.
- **BUILT** Switching to the no-registration path deletes uploaded certificates and signatory IDs; checks use only the declared business and e-KYC.
- **BUILT** Text is read on the device (OCR for photos, the text layer for PDFs); the OCR engine is served by our own app, not a CDN.
- **BUILT** The AI advisor never receives registry data or documents, and identifiers typed into the chat are removed first. The lead model uses profile fields only.

4 · CONSENT AND PURPOSE LIMITATION

- **BUILT** A plain notice in the owner's language before any check: what we check, why, and never for marketing.
- **BUILT** Consent is purpose-specific and recorded against the business and owner; every lookup and upload needs it. The 30-minute window only limits the session; fresh consent is asked when needed.
- **BUILT** Withdrawal stops further consent-based checks and is logged. Before verification, documents and identifiers are erased; after it, erasure requests are reviewed, and KYC a live service legally needs is kept only for that period.
- **AT LAUNCH** A registered DPDP consent manager; a separate marketing opt-in; approved WhatsApp templates.

5 · THIRD PARTIES AND TRANSFERS

- **Cloudflare R2**, **MongoDB**: processors under contract, in India or permitted regions.
- **UIDAI e-KYC**: via an authorised partner, with the owner's OTP consent each time.
- **Anthropic Claude** (advisor): screened chat text and profile buckets only; before launch, zero-retention terms and a permitted cross-border transfer.
- **Meta WhatsApp Cloud API**: processes phone numbers and whatever the owner types. No documents are asked for there; identifiers are removed before AI processing.

6 · OWNER RIGHTS AND ACCOUNTABILITY

- **BUILT** Each KYC check shows why it passed or failed; "ask a reviewer".
- **BUILT** My data: see consents and documents; ask for access, correction or erasure from the OTP-signed-in account, so the owner is identified. Requests get a due date. Grievance officer named on every page.
- **AT LAUNCH** Incident response with breach assessment and notification to the Data Protection Board; a DPIA; quarterly bias review of the lead model by language and city tier.

7 · RETENTION

Data	Today	At launch
Consent	30 min per grant; record kept	Consent evidence for the statutory period
One-time codes	Hashed; auto-expire	Same
KYC documents	Replaced files deleted; erased on withdrawal or request	Deleted after verification unless telecom KYC rules require it
Abandoned quotes	Kept, with a reminder after 24 h	Erased after 30 days
WhatsApp, leads, registry results, audit log, backups	No automatic expiry yet	A documented operational or legal period, then deleted or anonymised, backups included

All data is synthetic, from the case pack. A design note, not proof of compliance: confirm retention against DPDP rules and telecom KYC conditions.